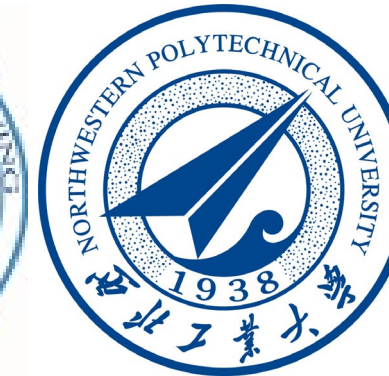
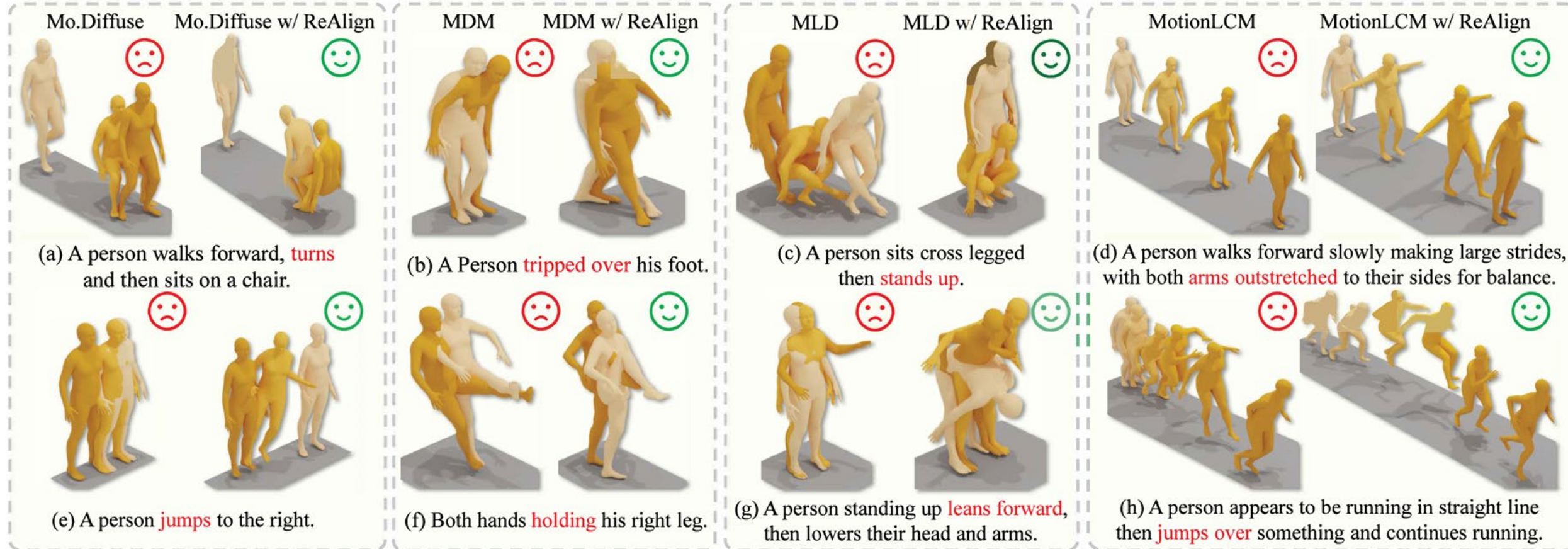




Introduction

Issue: Existing text-to-motion methods suffer from semantic misalignment and fail to rectify the denoising process.



Key Observation: We observe that diffusion-based text to motion generation accumulates text-motion misalignment during denoising.

Question: Can we improve text-motion alignment and realism at inference time?

Contributions: We propose ReAlign, a plug and play reward guided sampling method with a step aware reward model to improve text motion alignment and motion realism during denoising without fine tuning the diffusion model.

Motivation: Diffusion based text to motion models can generate realistic motions but often drift from the text because sampling favors high probability regions and image trained text encoders do not capture motion dynamics. Since paired motion text data is limited, we guide the sampling process with a reward distribution learned from available pairs to improve alignment without fine tuning the diffusion model.

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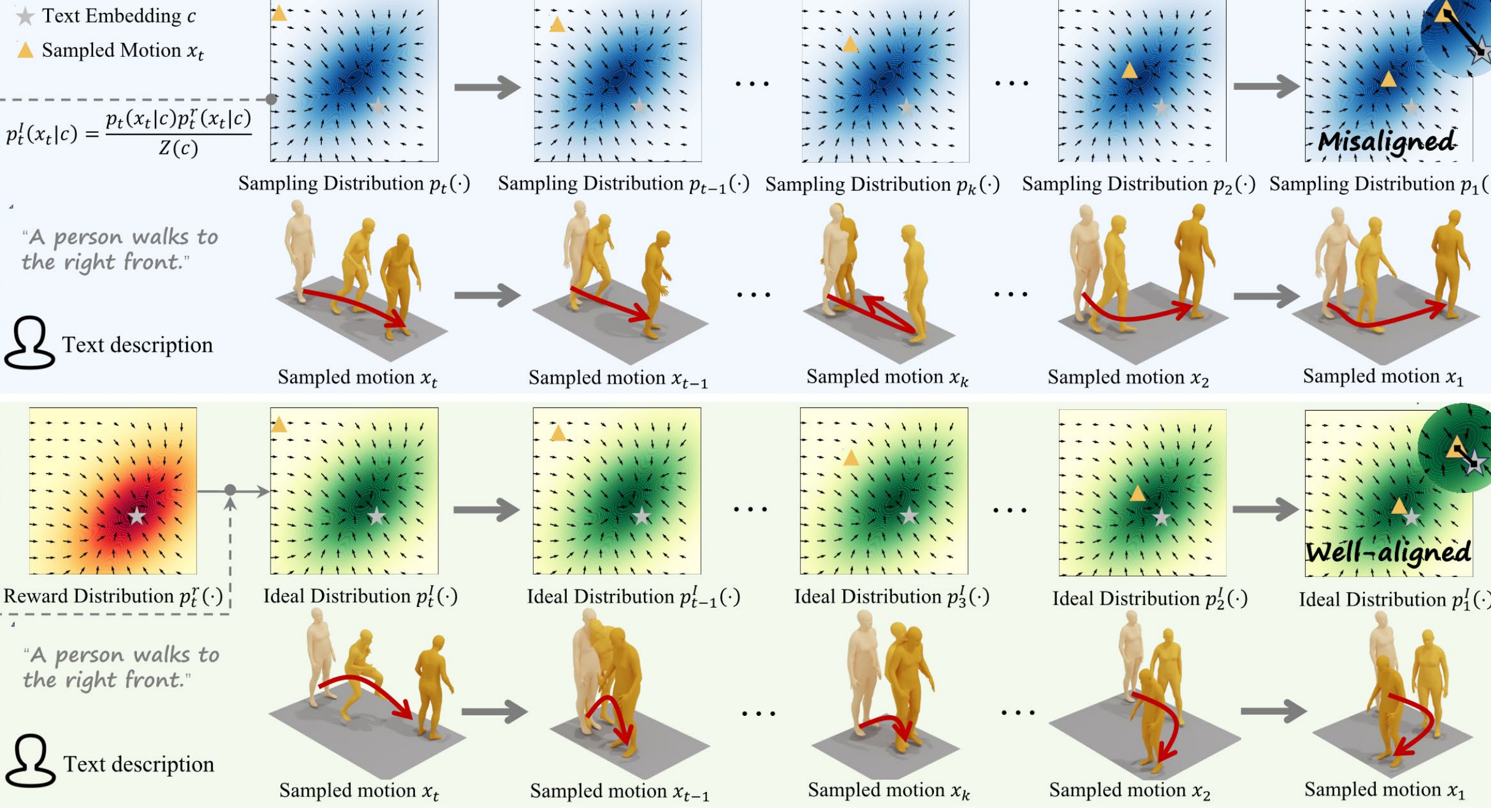


Code

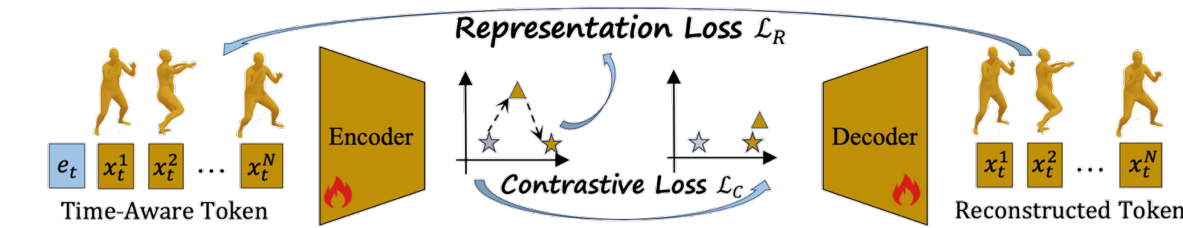


Paper

Toy Example



Reward-Guided Sampling



Algorithm 1: Training Step-Aware Reward Model

Input: Step-aware reward model R_φ , training set $\mathcal{D}_{tr}$, timestep T range $[t_{min}, t_{max}]$, probability parameter ω , noise scheduler α .

Output: Step-aware reward model R_φ .

- 1: **repeat**
- 2: **for** $(\mathbf{x}, c)$ in $\mathcal{D}_{tr}$ **do**
- 3: $t \leftarrow 0$ ▷ Initialize t
- 4: **if** $\text{Uniform}(0,1) > \omega$ **then**
- 5: $t \leftarrow \text{Uniform}(t_{min}, t_{max})$ ▷ Add noise to motion
- 6: **end if**
- 7: $\mathbf{x}_t \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}, (1 - \alpha_t)\mathbf{I})$ ▷ Forward process
- 8: $\mathcal{L}_{RM}(\varphi; \mathbf{x}_t, c) \leftarrow \mathcal{L}_C(\varphi; \mathbf{x}_t, c) + \mathcal{L}_R(\varphi; \mathbf{x}_t, c)$ ▷ Compute loss $\mathcal{L}_{RM}$
- 9: $\varphi \leftarrow \varphi - \nabla_\varphi \mathcal{L}_{RM}(\varphi)$ ▷ Update parameter
- 10: **end for**
- 11: **until** converged

Reward Distribution. With both the step-aware reward model and the motion-to-motion reward, we define the dual-alignment reward as:

$$R(\mathbf{x}_t, c) = \mu R_\varphi(\mathbf{x}_t, c) + \eta R_m(\mathbf{x}_t, c), \quad (9)$$

where μ and η control the contributions of text-based and

Algorithm 2: Reward-Guided Denoise Process

Input: Diffusion model ϵ_θ , reward model R , training set $\mathcal{D}_{tr}$, condition c , timestep T .

Output: Generated motion $\mathbf{x}_0$.

- 1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: $\mathbf{x}^c = \arg \max_{\mathbf{x} \in \mathcal{D}_{tr}} R_\varphi(\mathbf{x}, c)$
- 3: **for** $t = T, \dots, 1$ **do**
- 4: **use** $\mathbf{x}^c$ to obtain reward score
- 5: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ **if** $t > 1$ **else** $\epsilon = \mathbf{0}$
- 6: $R(\mathbf{x}_t, c) \leftarrow \mu R_\varphi(\mathbf{x}_t, c) + \eta R_m(\mathbf{x}_t, c)$ ▷ Compute Reward $R(\mathbf{x}_t, c)$
- 7: $\mathbf{x}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}} (\bar{\mathbf{x}}_{t-1} + \sqrt{\beta_t}\epsilon) + \nabla R(\mathbf{x}_t, c)$ ▷ Denoise guided by reward model
- 8: **end for**
- 9: **return** $\mathbf{x}_0$

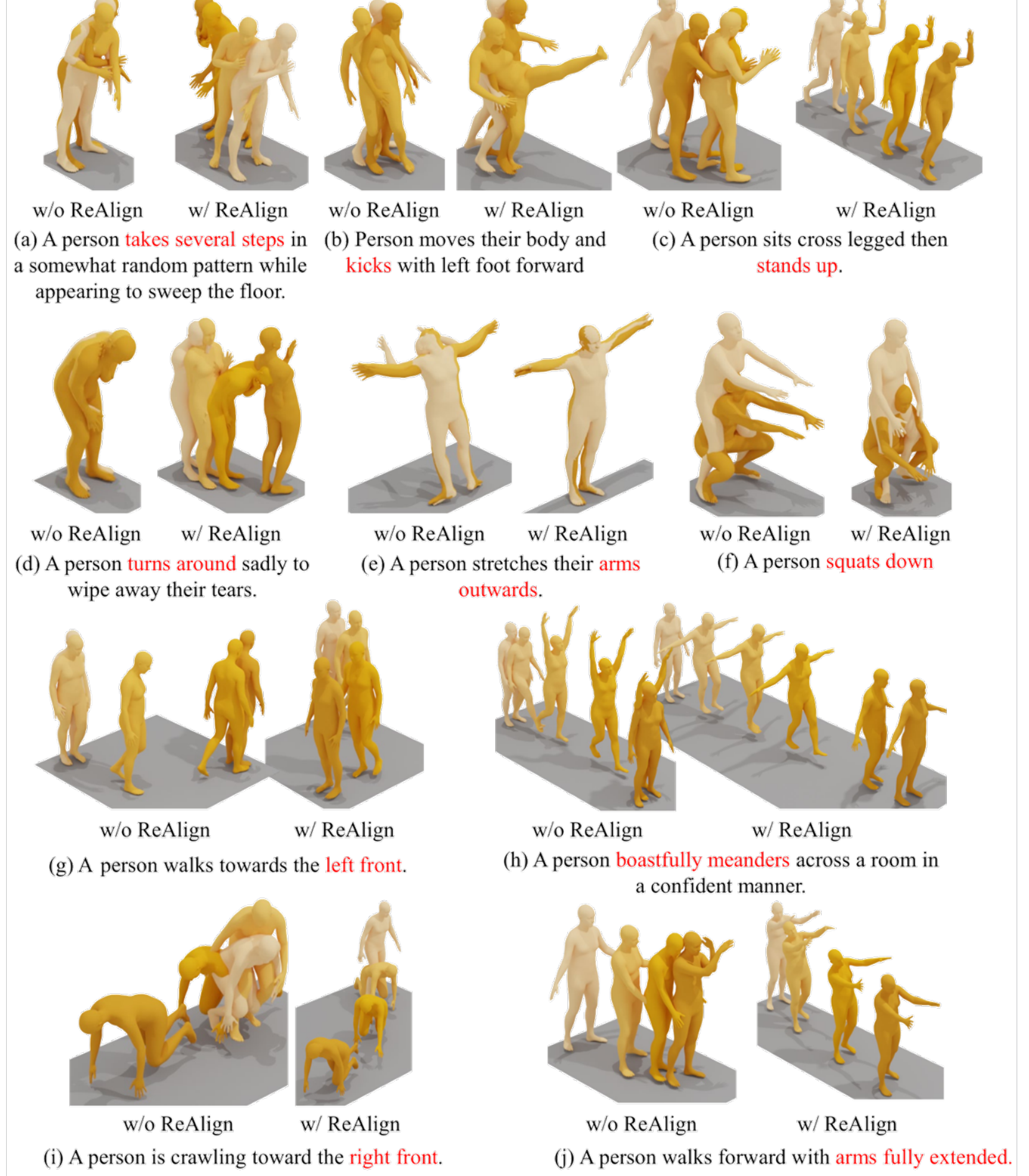
motion-based alignment. This reward formulation defines the reward distribution over noised motion as:

$$p_t^r(\mathbf{x}_t|c) = \exp(R(\mathbf{x}_t, c)) / Z^r(c). \quad (10)$$

Here, $Z^r(c) = \int \exp(R_\varphi(\mathbf{x}, c)) d\mathbf{x}$ is for normalization.

By integrating text-motion and motion-motion alignment, our approach constructs a robust reward signal that captures both semantic consistency and motion coherence. This enables more precise guidance of the diffusion sampling process, ensuring that generated motions are not only probable but also faithful to their textual descriptions.

Experiments



Method	R Precision $\uparrow$					FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$	
	Top 1	Top 2	Top 3	R@1	R@2				
Real	0.511	0.703	0.797	0.002	2.974	9.503			
T2M (2022a)	0.455 $\pm$ 0.002	0.636 $\pm$ 0.003	0.736 $\pm$ 0.003	1.087 $\pm$ 0.002	3.347 $\pm$ 0.008	9.175 $\pm$ 0.002			
MDM (2023)	0.455 $\pm$ 0.006	0.645 $\pm$ 0.007	0.749 $\pm$ 0.006	0.489 $\pm$ 0.047	3.330 $\pm$ 0.25	9.920 $\pm$ 0.083			
T2M-GPT (2023a)	0.492 $\pm$ 0.005	0.679 $\pm$ 0.006	0.775 $\pm$ 0.004	0.141 $\pm$ 0.009	9.722 $\pm$ 0.082	9.018 $\pm$ 0.075			
ReMoDiffuse (2023b)	0.510 $\pm$ 0.005	0.698 $\pm$ 0.006	0.795 $\pm$ 0.004	0.103 $\pm$ 0.004	2.973 $\pm$ 0.016	9.410 $\pm$ 0.049			
Mo.Diffuse (2024)	0.491 $\pm$ 0.001	0.681 $\pm$ 0.001	0.775 $\pm$ 0.001	0.630 $\pm$ 0.001	3.113 $\pm$ 0.001	9.657 $\pm$ 0.085			
OMG (2024)	-	-	0.784 $\pm$ 0.002	0.381 $\pm$ 0.008	-	9.607 $\pm$ 0.066			
MotionLCM (2025)	0.502 $\pm$ 0.003	0.698 $\pm$ 0.002	0.798 $\pm$ 0.002	0.304 $\pm$ 0.012	3.012 $\pm$ 0.007	9.871 $\pm$ 0.084			
Mamba (2025b)	0.502 $\pm$ 0.003	0.693 $\pm$ 0.002	0.792 $\pm$ 0.002	0.281 $\pm$ 0.011	3.060 $\pm$ 0.000	9.936 $\pm$ 0.060			
CoMo (2024)	0.502 $\pm$ 0.003	0.692 $\pm$ 0.002	0.790 $\pm$ 0.002	0.263 $\pm$ 0.011	3.032 $\pm$ 0.013	9.576 $\pm$ 0.073			
PuCo (2025)	0.515 $\pm$ 0.003	0.706 $\pm$ 0.003	0.801 $\pm$ 0.002	0.109 $\pm$ 0.005	2.927 $\pm$ 0.008	9.960 $\pm$ 0.073			
MARDM (2024)	0.500 $\pm$ 0.004	0.695 $\pm$ 0.003	0.795 $\pm$ 0.003	0.114 $\pm$ 0.007	-	9.500 $\pm$ 0.091			
MG-MotionLLM (2025b)	0.516 $\pm$ 0.002	0.706 $\pm$ 0.002	0.802 $\pm$ 0.003	0.303 $\pm$ 0.010	2.952 $\pm$ 0.009	9.724 $\pm$ 0.082			
EnergyMoGen (2025)	0.526 $\pm$ 0.003	0.718 $\pm$ 0.003	0.815 $\pm$ 0.002	0.176 $\pm$ 0.006	2.931 $\pm$ 0.007	9.474 $\pm$ 0.068			
MLD (2023b)	0.481 $\pm$ 0.003	0.673 $\pm$ 0.003	0.772 $\pm$ 0.002	0.473 $\pm$ 0.013	3.196 $\pm$ 0.010	9.474 $\pm$ 0.068			
w/ ReAlign (Ours)	0.567 $\pm$ 0.003	0.759 $\pm$ 0.003	0.848 $\pm$ 0.003	0.195 $\pm$ 0.005	2.704 $\pm$ 0.007	9.474 $\pm$ 0.068			
MLD++(2025)	0.548 $\pm$ 0.003	0.738 $\pm$ 0.003	0.829 $\pm$ 0.003	0.073 $\pm$ 0.003	2.810 $\pm$ 0.008	9.658 $\pm$ 0.089			
w/ ReAlign (Ours)	0.572 $\pm$ 0.002	0.764 $\pm$ 0.002	0.852 $\pm$ 0.001	0.055 $\pm$ 0.003	2.648 $\pm$ 0.008	9.478 $\pm$ 0.055			

Methods	Noise	Text-Motion Retrieval $\uparrow$					Motion-Text Retrieval $\uparrow$				
		R@1	R@2	R@3	R@5	R@10	R@1	R@2	R@3	R@5	R@10
HumanML3D	TEMOS (2022)	✗	40.49	53.52	61.14	70.96	84.15	39.96	53.49	61.79	72.40
	T2M (2022b)	✗	52.48	71.05	80.65	89.66	96.58	52.00	71.21	81.11	89.87
	T2M (2023)	✗	67.16	81.32	86.81	91.43	95.36	67.97	81.20	86.35	91.70
	LaMP (2025)	✗	67.18	81.90	87.04	92.00	95.73	68.02	82.10	87.32	92.20
	ReAlign (ours)	✓	67.59	82.24	87.44	91.97	96.28	68.94	82.86	87.95	92.44
KIT-ML	TEMOS (2022)	✗	43.88	58.25	67.00	74.00	84.75	41.88	55.88	65.62	75.25
	T2M (2022b)	✗	42.25	62.62	75.12	87.50	96.12	39.75	62.75	73.62	86.88
	TMR (2023)	✗	49.25	69.75	78.25	87.88	95.00	50.12	67.12	76.88	88.88
	ReAlign (ours)	✓	52.84	71.66	82.96	91.19	97.59	52.98	72.87	84.38	92.61